Malek Ahmad | majobs112002@gmail.com | +972549498961 | Linkedin |Github | Portfolio

Software Engineer specializing in **C/C++** and **Python**, with strong proficiency in **system-level programming**, **performance-oriented development**, and **Linux-based environments**. A **fast and self-driven learner** with **strong problem-solving skills**, **highly motivated team player**, and able to **work effectively under pressure**.

Skills

Programming languages: C, C++, Python, JavaScript.
IDEs: CLion, PyCharm, VS Code, Cursor.
Operating Systems: Linux (processes, memory, system calls), Windows.
Development Tools: Git, pytest, Valgrind.
System Programming: TCP/IP Socket Programming, Parallel Programming (MPI, OpenMP), Multithreading.
Database: SQL (MySQL), FireStore, MongoDB, SQLite.

Education

2021 –2025: Azrieli College of engineering| Software engineering| GPA 90.83

Courses: C/C++ Programming | Operating Systems | Data Structures | Algorithms | Object Oriented Programming | Parallel and distributed computation | Reinforcement Learning

Projects

- **2025: Retrieval-Augmented Generation (RAG) System for Multilingual Document Search,** Code link

Built a **multilingual document-search system** using **Flask, LangChain, and FAISS**, enabling **accurate Q&A** over large document collections.
Designed **backend architecture, vector indexing pipeline and REST APIs**, focusing on modularity and efficient data processing.

- **2025: Parallel Image Matching System,** Code link

Implemented a **hybrid parallel image-matching system** in **C** using **MPI** for distributed multi-process execution and **OpenMP** for intra-process thread parallelism, with **dynamic workload** distribution across cluster nodes.

- **2024: Indexed File System Simulator,** Code link

Implemented a **block-based file system simulator modeling inode structures** with **direct**, **single-indirect**, and **double-indirect allocation** using **C++**.
Designed **manual block management** using a **BitVector and performed low-level disk I/O operations** to simulate real storage behaviour.

- **2024: Concurrent Polynomial Processing System,** Code link

Designed a **multi-process producer–consumer** system in **C** using **System V shared memory** and **POSIX semaphores**.
Implemented **thread-level parallelism** with **pthreads** for polynomial operations, ensuring **synchronization and safe shared-memory management** in **Linux**.

Certificates

- **2025: Place-IL Developers Internship Program**
  - Successfully passed a **competitive** selection process (**250+ candidates**), including **coding challenges** in **OOP**, **data structures**, and **databases**.
  - Successfully completed **HR**, **technical**, and **group** interview stages.
- **2024: CS50: Introduction to Artificial Intelligence with Python – Harvard University (edX)** Projects and Certificate.
  - Completed an intensive course covering **search algorithms**, **optimization techniques**, **knowledge representation**, and **reinforcement learning**.
  - Implemented projects involving **algorithmic problem solving, constraint satisfaction, and performance-aware solutions** in **Python**.

Experience

2022 – 2025: Tutor | Azrieli College of Engineering | Jerusalem

Tutored **4-5 students per year** in **programming**, **algorithms**, **math** and **physics** courses, improving my responsibility, self-confidence, and socialization skills.

Languages

Arabic (Native) | Hebrew (Fluent) | English (High level)